Bayesian nonparametric glottal inverse filtering

The problem

Speech is locally expressible as the convolution of the glottal wave and the vocal tract filter.

We can measure speech directly, but not its constituents, the glottal wave and vocal tract filter.

Computing these constituents from speech alone is severely underdetermined. This fundamental problem is called speech deconvolution (or glottal inverse filtering) and has been open for 50+ years.

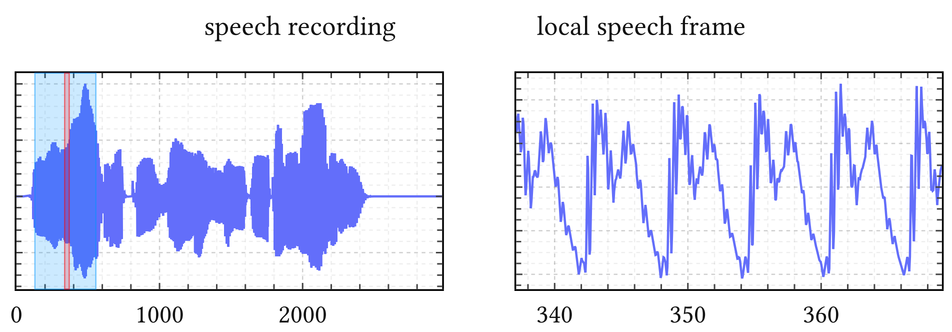
Next to its relevance to fundamental speech science, it is of practical importance for early diagnosis of neurodegenerative diseases like Parkinson's and Alzheimer's, because these interfere with speech production early on, as well as biomarking of heart failure.

Our solution

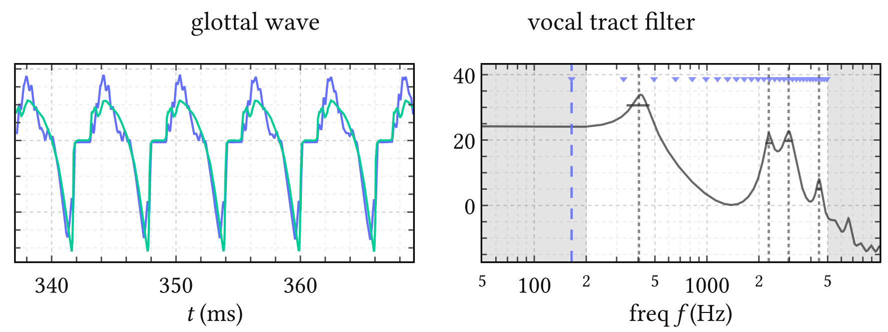
To handle the underdetermined nature of glottal inverse filtering, we recast the problem as Bayesian inference: Bayesian nonparametric glottal inverse filtering.

- We devise a novel method to learn a Gaussian process prior over glottal waveforms from a large corpus of expensive physics-based simulations.
- After embedding the corpus in a small latent space, we model it as a Gaussian mixture, and obtain a mixture of cheap surrogate models in data space.
- These surrogate models then define the prior for an efficient downstream variational inference method.
- Recent results show that our method achieves state of the art on the EGIFA speech benchmark, and cuts formant error on the VTRFormants benchmark by 30%.
- It is also fast for a fully probabilistic method: 5× realtime on a single GPU.

The data

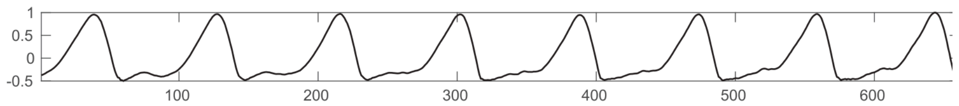


Source and filter

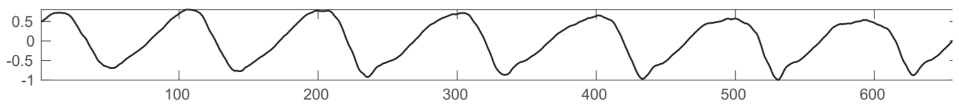


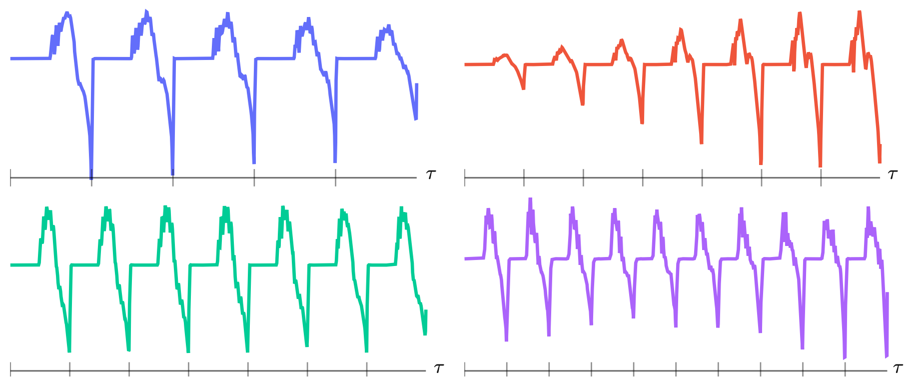
Speech biomarkers

glottal wave of healthy speaker



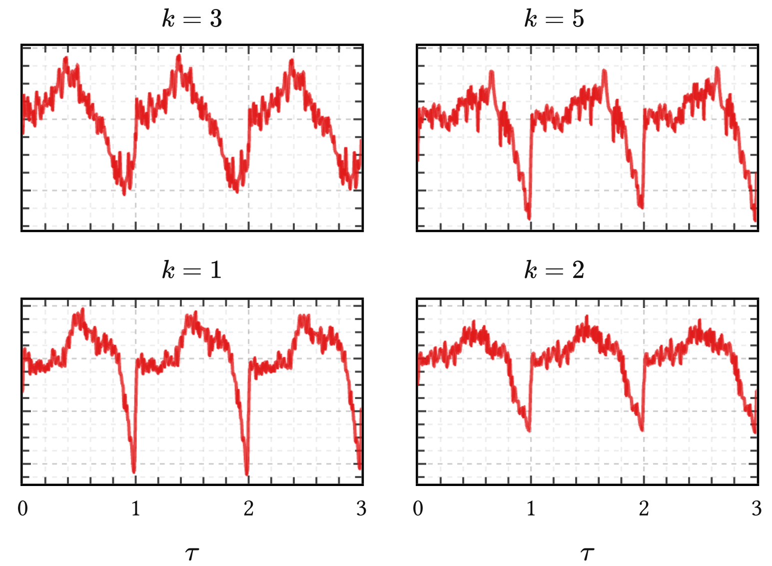
glottal wave of speaker with heart failure

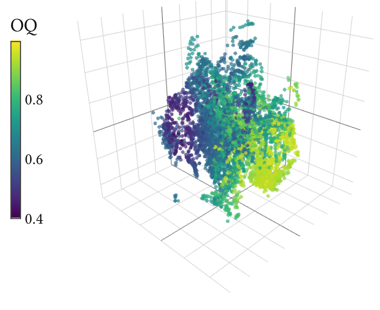




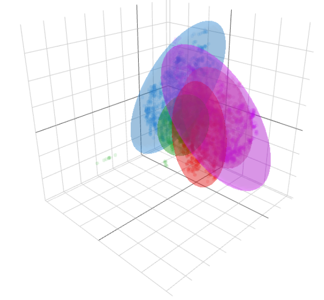
large corpus of glottal waves

Learned kernel bank





embedded corpus; each point is a glottal wave



mixture model of the embedded corpus